

Volume II · Chapter 9 — Medical Domain Adaptation

· Labs

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-II-Chapter-9-context.md for the AI interaction log.

Prerequisites: Ch 7, Ch 8.

Learning outcomes — the student can: distinguish continued pretraining vs. fine-tuning vs. instruction tuning; explain how Meditron/Clinical-Camel/MedGemma were adapted; choose an adaptation strategy; reason about data requirements and licensing.

Labs (hands-on) — 6 h:

#	Lab	h
9a	Compare a general vs. domain-adapted model on medical prompts	2
9b	Prepare a small biomedical instruction dataset (format, clean)	2
9c	Design an adaptation plan for a chosen clinical use case	2

Datasets/tools: HF models (Meditron/MedGemma/PubMedBERT), datasets.

Assessment: adaptation-plan deliverable (**50%**); comparison lab (**30%**); quiz (**20%**).

Key decisions: pretrain vs. fine-tune vs. RAG; data licensing; base-model choice.

References: plan §13 → *Medical LLMs; Fine-tuning, alignment & benchmarks*. **Hours:**

Theory **10** + Lab **6** = **16**.